KUMARAGURU COLLEGE OF TECHNOLOGY

Department of Artificial Intelligence and Data Science

DATA SCIENCE AND VISUALIZATION (24ADI204)

WEEK 4 REPORT

Team Formation & Data Hunting

Team 16: Youth Smoking and Drug Consumption Analysis

DATA-VIZ HACK-A-BIT 2026–2027

Team Member Name	Roll Number
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EDA Deep Dive – Statistical Analysis and Visualization

1. Week 4 Contribution

Performed Exploratory Data Analysis (EDA) and studied descriptive statistical measures to identify patterns and distributions in the dataset.

2. Work Done:

During Week 4, my individual work focused on performing Exploratory Data Analysis on the Youth Smoking and Drug Consumption dataset. I studied the statistical characteristics of numerical variables and explored the distribution of different features. Descriptive statistical measures were used to understand the central tendency, spread and shape of the data. I also explored visualizations to identify patterns and relationships between different variables.

3. Topics Studied

- **Mean, Median and Mode – Calculated measures of central tendency to understand the average, middle and most frequently occurring values in the numerical features.**
- **Variance and Standard Deviation – Analysed the spread of numerical values to understand the level of variation within the dataset.**
- **Skewness Analysis – Examined the asymmetry of numerical distributions to identify whether the data is positively skewed, negatively skewed or approximately symmetrical.**

- **Kurtosis Analysis – Studied the shape and tail characteristics of distributions to understand the presence of extreme observations.**
- **Univariate Analysis – Analysed individual variables using Histograms and Boxplots to understand their distributions and identify unusual patterns.**
- **Bivariate Analysis – Explored relationships between pairs of variables using Scatter Plots and other suitable graphical techniques.**
- **Statistical Visualization – Used Python visualization libraries such as Matplotlib and Seaborn to represent statistical findings graphically.**

4. Outcome of My Contribution

Through this analysis, I gained practical understanding of descriptive statistics and Exploratory Data Analysis. I learned how statistical measures and visualizations help identify patterns, variations and relationships within the Youth Smoking and Drug Consumption dataset. The analysis provided useful insights for further feature transformation and advanced statistical analysis.

5. Reference

Project notebook: DSV Team 16 – Youth Smoking and Drug Consumption Analysis.